TREDENCE AI GRADS · PROJECT EVALUATION

Participant_01

Milestone 1 · Retail Customer Purchase
Submission:
Submission-Milestone1/Participant_M1_01_retail_purchase_strong.ipynb
Evaluated: 27 Oct 2026

87 / 100
Distinction · 87%

Criterion-wise marks

Data Understanding & Quality Checks — 18 / 20

Strong data loading and join validation with missingness and duplicates checked. Could improve by deeper schema inspection (e.g., unique keys in demographics, purchase) and more detailed missingness patterns or data type validations.

Exploratory Data Analysis (EDA) — 17 / 20

Good coverage of target distribution and categorical group differences with visual and tabular summaries. Could improve by including outlier detection, bivariate plots, or more detailed pattern interpretations.

Data Cleaning & Feature Engineering — 16 / 20

Appropriate missing value treatment and encoding with leakage prevention. No explicit new feature engineering beyond encoding. Could improve by justifying imputation choices or exploring feature creation from existing data.

Linear Model Building — 19 / 20

Well-structured baseline model with proper data splitting and pipeline usage. Could improve by documenting assumptions about linearity or feature relevance more explicitly.

Evaluation & Findings — 17 / 20

Evaluation metrics are correctly reported and limitations acknowledged. Could improve by interpreting error magnitudes in business context and discussing model limitations more thoroughly.

Strengths

• Comprehensive data loading and join validation with missingness and duplicates checked.
• Clear and appropriate exploratory data analysis with visualizations and group summaries.
• Proper data cleaning and preprocessing pipeline with leakage prevention.
• Reproducible baseline linear regression model with train/test split and pipeline.
• Correct reporting of evaluation metrics and cautious interpretation without unsupported claims.

To improve next time

• Deeper schema inspection and detailed missingness pattern analysis.
• More extensive EDA including outlier detection and bivariate relationships.
• Justification and exploration of feature engineering beyond encoding.
• Explicit documentation of modeling assumptions and feature relevance.
• More detailed interpretation of evaluation metrics in business terms and model limitations.

Overall feedback

The submission meets the milestone 1 requirements well, demonstrating solid data understanding, exploratory analysis, cleaning, baseline modeling, and evaluation. It avoids common pitfalls such as leakage and unsupported claims. To strengthen the work, deeper data quality insights, richer EDA, and more thorough interpretation of results and limitations are recommended. The notebook is well organized and reproducible, supporting human review.

AI-generated evaluation against the milestone rubric. Subject to trainer review. Each submission is evaluated once; this score is recorded.